

Metadata

Title

Island-Based Differences in Size Measurements of Female Adelie Penguins of the Palmer Archipelago

Description

This observational study seeks to investigate the size differences in female members of the Adelie Penguin (*Pygoscelis adeliae*) species within the islands of the Palmer Archipelago. Specifically, this study aims to analyze the difference in flipper length (mm), and body mass (g) between the female members of the Adelie species within the islands of the Palmer Archipelago: Biscoe, Dream and Torgerson.

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Study Information

Research questions

Research Question 1: Is there a difference in flipper length (mm) of female Adelie Penguins between the Biscoe, Dream and Torgerson islands of the Palmer Archipelago.

Research Question 2: Is there a difference in body mass (g) of female Adelie Penguins between the Biscoe, Dream and Torgerson islands of the Palmer Archipelago.

Hypotheses

Hypothesis 1 (non-directional): There is a mean difference in the bill length (mm) of female Adelie Penguins between the Biscoe, Dream and Torgerson islands of the Palmer Archipelago ($\mu_B \neq \mu_D \neq \mu_T$).

Hypothesis 2 (non-directional): There is a mean difference in the body mass (g) of female Adelie Penguins between the Biscoe, Dream and Torgerson islands of the Palmer Archipelago ($\mu_B \neq \mu_D \neq \mu_T$).

Data Description

Datasets used

The dataset being used in this study was acquired from *structural size measurements and isotopic signatures of foraging among adult male and female Adélie penguins (Pygoscelis adeliae) nesting along the Palmer Archipelago near Palmer Station* by the Palmer Station Antarctica LTER in 2020, and can be accessed [here](#). In the study, the data was collected seasonally (longitudinal) to examine ecological sexual dimorphism between Pygoscelis penguins as a result of foraging variability caused by changing levels of winter sea ice¹.

Data availability

All data and code are publicly available for this study and can be reached at [this](#) github repository.

Access date

The data for this project was accessed from the study above on 2026-04-09.

Data collection procedures

Detailed data collection methods can be viewed at the Palmer Station Antarctica LTER website or accessed [here](#).

Variables

Measured variables

The explanatory variable in this study is the island within the Palmer archipelago where the female Adélie penguins were observed: Biscoe, Dream and Torgerson.

The two response variables in this study are body mass (g) and flipper length (mm) of the female Adélie penguins.

Missing data

There is no missing data in the datasets associated with body mass (g) or flipper length (mm) of any of the female Adélie penguins observed in this study.

Unit of analysis

The original dataset will be restricted to only include female members of the Adelie penguin species. The dataset will then be processed to only include the variables: Sex, Species, Region, Penguin ID, Island, Body mass and Flipper length.

Statistical outliers

Statistical outliers will not be removed.

Sampling weights

No sampling weights are available for this dataset.

Knowledge of Data

Prior Publication/Dissemination

There has been no prior publications, working papers or conference presentations by the author on the data used in this study.

Prior knowledge

The author of this study is previously aware of the data used in this study and has previously conducted analyses of a modified version of the dataset.

Analyses

Statistical models

No statistical tests will be performed in this study. The data will only be used to visually assess the associations between the body mass (g) and flipper length (mm) using violin plots.

Effect size

This study will likely have a small effect size as there are probably few environmental differences between the islands within the Palmer archipelago. Meaning, the foraging patterns between the penguins are likely similar and therefore their size measurements may also be similar.

Statistical power

This study will likely have low-moderate statistical power as the effect size will be small but will have decent sample sizes between the islands: Biscoe (n=22), Dream (n=27) and Torgerson (n=24) Islands.

Inference criteria

Given that no statistical tests will be performed, all inferences made from the data will be purely visual from the violin plots that were generated.

Assumption Violation/Model Non-Convergence

There is no need to check assumptions for statistical tests as none will be used.

References

- (1) LTER, P. S. A.; Gorman, K. Structural Size Measurements and Isotopic Signatures of Foraging among Adult Male and Female Adélie Penguins (*Pygoscelis Adeliae*) Nesting along the Palmer Archipelago near Palmer Station, 2007-2009, 2020.
<https://doi.org/10.6073/PASTA/98B16D7D563F265CB52372C8CA99E60F>.